

Tiger Lu

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APPOINTMENTS	Center for Computational Astrophysics , Flatiron Institute, New York, NY Flatiron Research Fellow	Sep 2025 – Present
	Yale University , New Haven, CT Graduate Research Assistant	Aug 2020 – May 2025
	University of California, Berkeley , Berkeley, CA Summer Undergraduate Researcher	May 2019 – Aug 2019
	Carnegie Observatories , Pasadena, CA Summer Undergraduate Researcher	May 2018 – Aug 2018

EDUCATION	Yale University , New Haven, Connecticut, USA	
	- Ph.D. in Astronomy with Distinction • Thesis: Planets are not Points: the Profound Effect of Planetary Structure on Exoplanet System Architectures • Advisers: Prof. Gregory Laughlin, Prof. Malena Rice	Aug 2020 – May 2025
	M.S., M.Phil. in Astronomy	Aug 2020 – Aug 2022
	California Institute of Technology , Pasadena, California, USA	
	- B.S. in Astrophysics • Minor in Computer Science	Aug 2016 – Jun 2020

PUBLICATIONS 193 citations. h -index = 8.
Italics: Student directly supervised by T. Lu

FIRST-AUTHOR

- [9] **Lu, T.**, Hernandez, D.M., Rice, M. and Su, Y. In review.
- [8] **Lu, T.** and Brown, Garrett. [A Coordinate System for Dynamical Instabilities in Hierarchical Systems in REBOUND](#). In review.
- [7] **Lu, T.**, Millholland, S.C., Rice, M., Black, B., Bardalez Gagliuffi, D.C., Balmer, W.O., Pueyo, L., Giovinazzi, M.R. and Brandt, T.D. [The Architecture of the 14 Herculis System Suggests Primordial Ejection of a Massive Planet](#). *ApJ* in press.
- [6] **Lu, T.**, Tajer, H., Hernandez, D.M., Rein, H., *Liu, Y.* and Rice, M., 2025. [Collisional Fragmentation Support in TRACE](#). *RNAAS* **9**, 5
- [5] **Lu, T.**, Li, G., Cassese, B. and Lin, D.N.C., 2025. [The Dynamical History of HIP-41378 f - Oblique Exorings Masquerading as a Puffy Planet](#). *ApJ* **980**, 39
- [4] **Lu, T.**, An, Q., Li, G., Millholland, S.C., Rice, M., Brandt, G.M. and Brandt, T.D., 2025. [Planet-Planet Scattering and ZLK Migration – the Dynamical History of HAT-P-11](#). *ApJ* **979**, 218
- [3] **Lu, T.**, Hernandez, D.M. and Rein, H., 2024. [TRACE: a Time-Reversible Algorithm for Close Encounters](#). *MNRAS* **533**, 3708
- [2] **Lu, T.**, Rein, H., Tamayo, D., Hadden, S., Mardling, R., Millholland, S.C. and Laughlin, G., 2023. [Self-Consistent Spin, Tidal and Dynamical Equations of Motion in the REBOUNDx Framework](#). *ApJ* **948**, 41
- [1] **Lu, T.** and Laughlin, G., 2022. [Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9](#). *PSJ* **3**, 221

STUDENT-LEAD

- [4] *Yang, E.*, **Lu, T.**, Yahalomi, D.A. and Winn, J.N., 2026. [A Massive Hot-Jupiter Companion that Disfavors Giant Planet Formation Beyond the Water-Ice Line](#). *ApJL* **1004**, L29
- [3] *Im, H.*, **Lu, T.**, Rice, M., Tran, Q., Li, G. and Naoz, S., 2026. [Observational and Dynamical Constraints on an Unseen Outer Perturber in the GJ 436 Hot Neptune System](#). *ApJ* **1003**, 84

- [2] *Liu, Y., Lu, T. and Rice, M., 2026. Double Hot Jupiter Formation through Mirrored ZLK Migration in Binary Star Systems: The Case of WASP-94. ApJL 997, L41*
- [1] *Liu, Y., Lu, T. and Rice, M., 2025. The Formation of Double Hot Jupiter Systems Through ZLK Migration. ApJ 986, 103*

SIGNIFICANT CONTRIBUTION

- [6] Rescigno, F., Stalport, M., Anna John, A., **Lu, T.**, et al. 2026. [Understanding eccentric temperate giants: an in-depth study of the architecture and stellar obliquity of the TOI-2134 system.](#) *MNRAS* In press.
- [5] Yahalomi, D.A., **Lu, T.**, Armitage, P., Bedell, M., Casey, A.R., Price-Whelan, A. and Rice, M., 2026. [The Astrometric Resoeccentric Degeneracy: Eccentric Single Planets Mimic 2:1 Resonant Planet Pairs in Astrometry.](#) *ApJL 999, L9*
- [4] Cassese, B., Rice, M. and **Lu, T.**, 2025. [A High-Precision, Differentiable Code for Solar System Ephemerides.](#) *PSJ 6, 252*
- [3] An, Q., **Lu, T.**, Brandt, G.M., Brandt, T.D. and Li, G., 2025. [Significant Mutual Inclinations Between the Stellar Spin and the Orbits of Both Planets in the HAT-P-11 System.](#) *AJ 169, 22*
- [2] Cassese, B., Vega, J., **Lu, T.**, Rice, M., Poddar, A. and Kipping, D., 2024. [squishyplanet: Modeling transits of non-spherical planets in JAX.](#) *JOSS 9, 100*
- [1] Wahl, S., Thorngren, D., **Lu, T.** and Militzer, B., 2021. [Tidal Response and Shape of Hot Jupiters](#) *ApJ 921, 105*

OTHER CO-AUTHOR

- [4] Naponiello, L., Poretti, E., Rice, K., et al. (inc. **Lu, T.**) in review.
- [3] Tajer, H., Wang, J., Childs, A., et al. (inc. **Lu, T.**) 2026. [Origins of Mercury’s Big Heart of Iron: Exploring Pathways to Form High Core Mass Fraction \(CMF\) Planets via N-body Simulations.](#) *ApJ 998, 46*
- [2] Bardalez Gagliuffi, D.C.†, Balmer, W.O.†, Pueyo, L., et al. (inc. **Lu, T.**) 2025. [JWST Coronagraphic Images of 14 Her c: a Cold Giant Planet in a Dynamically Hot, Multi-planet System.](#) *ApJL 988, L18*
- [1] Wang, N., Lu, L.Y., Liu, H.G., et al. (inc. **Lu, T.**) 2025. [Development of Gravity Theories in the View of TRAPPIST-1e.](#) *RAA 25, 015003*

WHITEPAPERS

- [1] Levine, W. G., Gerbig, K., Loudén, E., **Lu, T.**, Hsieh, C., O’Connor, C., Li, R. and Dong, J., 2024. [Emerging Researchers in Exoplanetary Science \(ERES\): Lessons Learned in Conference Organization for Early-Career Researchers.](#) *Bulletin of the AAS 56, 2024i004*

†: Joint first author

AWARDS & FELLOWSHIPS

▪ Yale Dirk Brouwer Prize for Outstanding Thesis of Unusual Merit	2026
▪ DDA/AAS Raynor L. Duncombe Student Research Prize	2025
▪ Flatiron Research Fellowship	2025
▪ Flipped Science Fair Best Poster Explanation	2024
▪ AAS International Travel Grant	2024
▪ Graduate Student Assembly Conference Travel Fellowship	2024
▪ National Science Foundation GRFP (<i>Honorable Mention</i>)	2022
▪ Joan and Arnold Seidel Griffith Observer Science Writing Contest (<i>Second Place</i>)	2022
▪ Alan Porter Memorial Foundation Fellow	2019
▪ Summer Undergraduate Research Fellow	2018

ORAL PRESENTATIONS & TALKS

▪ New York Area Exoplanets Meeting III, AMNH (<i>New York City, NY</i>)	May 2026
▪ Monday Afternoon Talk, MIT (<i>Cambridge, MA</i>)	Apr 2026
▪ Astronomy & Planetary Science Colloquium*, NAU (<i>Flagstaff, AZ</i>)	Feb 2026
▪ Astronomy Seminar*, AMNH (<i>New York City, NY</i>)	Feb 2026
▪ International Conference on Exoplanets and Planet Formation (<i>Shanghai, China</i>)	Dec 2025
▪ Powell Group Meeting, University of Chicago (<i>Chicago, IL</i>)	Oct 2025
▪ 246th AAS Meeting Dissertation Talk (<i>Anchorage, AK</i>)	Jun 2025
▪ Boston Area Planetary Science Meeting, MIT (<i>Cambridge, MA</i>)	May 2025

▪ New York Area Exoplanets Meeting II, CCA (<i>New York City, NY</i>)	May 2025
▪ 56th DDA Meeting, Georgia Institute of Technology (<i>Atlanta, GA</i>)	May 2025
▪ ML in Planetary Dynamics Workshop*, CCA (<i>New York City, NY</i>)	Apr 2025
▪ Tea Talk, Indiana University (<i>Bloomington, IN</i>)	Feb 2025
▪ ITC Luncheon, Harvard University (<i>Cambridge, MA</i>)	Jan 2025
▪ Exoplanet Pizza Lunch, Harvard University (<i>Cambridge, MA</i>)	Jan 2025
▪ Exoplanet Seminar, Princeton University (<i>Princeton, NJ</i>)	Nov 2024
▪ Planetary Lunch Seminar, Massachusetts Institute of Technology (<i>Cambridge, MA</i>)	Nov 2024
▪ Astronomy Seminar, Columbia University (<i>New York City, NY</i>)	Oct 2024
▪ Astronomy Seminar, Stony Brook University (<i>Stony Brook, NY</i>)	Oct 2024
▪ CRA Seminar, Georgia Institute of Technology (<i>Atlanta, GA</i>)	Oct 2024
▪ Planetary Science Seminar, California Institute of Technology (<i>Pasadena, CA</i>)	Oct 2024
▪ Lunch Seminar, UCLA (<i>Los Angeles, CA</i>)	Sep 2024
▪ Exoplanet Group Meeting, UCSB (<i>Santa Barbara, CA</i>)	Sep 2024
▪ FLASH Talk, UCSC (<i>Santa Cruz, CA</i>)	Sep 2024
▪ Star and Planet Lunch, University of Hawai'i (<i>Honolulu, HI</i>)	Aug 2024
▪ CT Exoplanet Picnic, Wesleyan College (<i>Wesleyan, CT</i>)	Jul 2024
▪ Emerging Researchers in Exoplanet Science IX, Cornell University (<i>Ithaca, NY</i>)	Jul 2024
▪ REBOUND 24 Meeting* (<i>Virtual</i>)	Jul 2024
Arbitrary ODE Workshop	
▪ REBOUND 24 Meeting* (<i>Virtual</i>)	Jul 2024
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▪ New York Area Exoplanets Meeting I, CCA (<i>New York City, NY</i>)	May 2024
▪ 55th DDA Meeting, University of Toronto (<i>Toronto, Canada</i>)	May 2024
▪ Makino Lab Seminar, Kobe University (<i>Kobe, Japan</i>)	Jul 2023
▪ Exoplanet Seminar, Nanjing University (<i>Nanjing, China</i>)	Jul 2023
▪ 54th DDA Meeting, Michigan State University (<i>East Lansing, MI</i>)	May 2023
▪ Exoplanet Seminar, Princeton University (<i>Princeton, NJ</i>)	Mar 2023
▪ Astronomy Seminar, Monash University (<i>Monash, Australia</i>)	Feb 2023
▪ 240th AAS Meeting (<i>Pasadena, CA</i>)	Jun 2022

*Invited Talk

POSTER PRESENTATIONS

▪ 57th DDA Meeting, University of Chicago (<i>Chicago, IL</i>)	Jun 2026
▪ Emerging Researchers in Exoplanet Science, Princeton University (<i>Princeton, NJ</i>)	Jun 2025
▪ Exoplanets V Meeting (<i>Leiden, The Netherlands</i>)	Jun 2024
▪ Extreme Solar Systems (<i>Christchurch, New Zealand</i>)	Jun 2024
▪ 2022 AGU Meeting (<i>Chicago, IL</i>)	Dec 2022
▪ Exoplanets IV Meeting (<i>Las Vegas, NV</i>)	May 2022
▪ 2019 AGU Meeting (<i>San Francisco, CA</i>)	Dec 2019
▪ 233rd AAS Meeting (<i>Seattle, Washington</i>)	Jan 2019

TEACHING

ASTR 210: Stars and Their Evolution	New Haven, CT, USA
Yale University Teaching Fellow	Spring 2023
ASTR 160: Frontiers and Controversies in Astrophysics	New Haven, CT, USA
Yale University Teaching Fellow	Fall 2022
ASTR 160: Frontiers and Controversies in Astrophysics	New Haven, CT, USA
Yale University Teaching Fellow	Spring 2021
ASTR 155: Introduction to Astronomical Observing	New Haven, CT, USA
Yale University Teaching Fellow	Fall 2021
Dean's Tutor	Pasadena, CA, USA
Tutoring Caltech students in Physics, Astronomy and Computer Science	2017-2020

OUTREACH

Warrior-Scholar Program	New Haven, CT, USA
Research Project Leader	Summer 2023
<i>One week intensive boot camp preparing military veterans for undergraduate degrees</i>	

Yale OpenLabs

Local outreach organization dedicated to educating middle school children on STEM topics. Receptient of the Seton Elm-Ivy Award.

- Public Relations Officer 2023
- Regular Outreach Speaker 2022 – 2025
 - Science Cafe 2024 Speaker: *What if 'Cotton Candy' Planet are Actually Ring Pops?*
 - Flipped Science Fair Best Poster Explanation: *Probing the Shapes of Planets*
 - Science Cafe 2022 Speaker: [How We Discover Exoplanets](#)

Astronomy on Tap

New Haven branch of outreach program aimed at engaging local community with astronomy research

- Social Media Director 2022

Science in the News

Local outreach organization dedicated to educating the greater community on STEM topics.

- Outreach Speaker 2021
 - [Hidden Things: Pulling Back the Veil on the Universe](#)

RESEARCH MENTORING

Yurou (Nina) Liu*

Jun 2023 –

Alexander P. Hixon Summer Undergraduate Research Fellow

Yale University

Project: *The Formation of Double Hot Jupiter Systems Through ZLK Migration*

Haedam Im

May 2025 – Apr 2026

Dorrit Hoffleit Undergraduate Research Fellow

Massachusetts Institute of Technology

Project: *Hunting for a Hidden Planet in the GJ 436 System*

Vighnesh Nagpal

Aug 2025 –

Graduate Student

University of Chicago

Haniyeh Tajer

May 2025 –

Graduate Student

Ohio State University

Eritas Yang

Oct 2025 – Apr 2026

Graduate Student

Princeton University

Project: *A Massive Hot Jupiter Companion Disfavors Formation Beyond the Ice Line*

Jeremiah Reynoso

Jun 2023 – Oct 2023

Summer Undergraduate Research Fellow

Morehouse College

*Primary Research Advisor

INFORMAL MENTORING

Chris Santiago

2022 – 2024

Astro Sibb

Yale University

Kylyn Smith

2023 – 2024

Astro Sibb

Yale University

Sally Jiang (current Ph.D. Candidate, Columbia University)

2022 – 2023

Astro Sibb

Yale University

TELESCOPE PROPOSALS (CO-I)

James Webb Space Telescope – 44.7 hours

Cycle 5 GO

ID: 11253. Fluffy or Flashy? Resolving the Mystery of the Coldest-Known Super Puff. *PI: Ben Cassese*

Keck Observatory (KPF) – 0.5 nights

Yale 2026A

Directly Probing the Migration History of WASP-53 with Precise Orbital Constraints. *PI: Quang Tran*

Keck Observatory (KPF) – 2 nights

Yale 2024B

The Search for Double Hot Jupiter Systems. *PI: Malena Rice*

Keck Observatory (HIRES) – 2 nights

Yale 2023A

Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 2). *PI: Emma Louden*

Keck Observatory (HIRES) – 2 nights

Yale 2022B

Probing the Exoplanet Mass Discrepancy Between the Radial Velocity and Transit Timing Methods with the Anomalously Low Density Planet Sample and Keck-HIRES (Phase 1). *PI: Emma Louden*

PROFESSIONAL SERVICE	Science Organizing Committee – 57th DDA Meeting <i>Division of Dynamical Astronomy, University of Chicago</i>	June 2026
	Science Organizing Committee Chair – DSPS <i>A Dynamical Synthesis of Planetary Systems, New Haven</i>	Jan 2026
	Organizing Committee – ERES X <i>Emerging Researchers in Exoplanet Science Symposium, Princeton University</i>	July 2025
	Science Organizing Committee – ERES VIII <i>Emerging Researchers in Exoplanet Science Symposium, Yale University</i>	June 2023
	Reviewer <i>NASA XRP, NASA Solar System Workings Panel, A&A, AJ, ApJ, MNRAS</i>	2024 – Present
	Main Co-Organizer – Yale Exoplanets and Stars Seminar <i>Weekly Seminar Series</i>	2023 – 2025
	Committee Member – ACDC <i>Yale University Astronomy Climate and Diversity Committee</i>	2022 – 2025
MEDIA	Yale Daily News New study offers a double dose of ‘hot Jupiters’	Jun 2025
	Popular Science What if ‘cotton candy’ planets are actually Ring Pops?	Oct 2024
	AAS Author Journal Series Tilting Uranus via Secular Spin-Orbit Resonance with Planet 9	Nov 2022
COMMUNITY ENGAGEMENT	New Haven Road Runners – New Haven, CT, USA <i>Local USATF sanctioned running club</i>	
	▪ Public Relations Officer	Jul 2023 – Jul 2024
	▪ Racing Team Director / Elite Athlete Coordinator	Jul 2022 – Jun 2025
	NCAA Athlete – Pasadena, CA, USA <i>NCAA Division III Cross Country/Track & Field at Caltech</i>	2016 – 2020
	Avery House – Pasadena, CA, USA <i>Student Government</i>	
	▪ Athletics Manager	2017 – 2018
	▪ Underclassmen Counselor	2017 – 2020

[CV compiled on 2026-07-02]